VisionGuard: AI vs. Real Image Classification

End-to-End MLOps Pipeline, Experiment Tracking, Containerization & Cloud Deployment

Project Identifier: 23L0713_A2_Ops | FAST-NUCES MLOps Assignment 2

Front Matter -- Student, Repository & Deployment Metadata

Field	Value
Student Name	Abdullah Chouhan
Roll Number	23L0713
Project Name (Everywhe	23L0713_A2_Ops
GitHub Repository	https://github.com/Abd
DagsHub Repository	https://dagshub.com/Ab
MLflow Tracking URI	https://dagshub.com/Ab
Live Cloud API URL	https://abdullahchouha
Graded Git Commit Hash	7dd53376307569b5
Generated Timestamp	2026-11-05 03:25:24 PK

Chapter 1 -- Task 1: Data Understanding and Versioning (7 Marks)

Task 1.1: Isolated Python Environment & Hardware Verification

What We Did: Created an isolated Python 3.11.18 virtual environment (.venv) using uv (uv venv --python 3.11 .venv) and installed pinned versions of TensorFlow (2.18.0), Keras (3.15.1), Keras-Tuner (1.4.7), MLflow (2.19.0), DagsHub (0.5.9), DVC (3.58.0) with dvc-s3 (3.2.0), Flask (3.1.0), SQLAlchemy (2.0.36), Pillow (11.0.0), Scikit-Learn (1.6.0), and Pytest (8.3.4).

Why & How It Works: Isolating dependencies in a virtual environment prevents version collisions across host packages and guarantees deterministic builds between local development, GitHub Actions CI, and Docker containers. We wrote src/data/env_check.py using importlib.metadata and tf.config.list_physical_devices('GPU') to record exact versions and verify hardware acceleration.

What Went Wrong & Fix: The host system defaulted to Python 3.14.7, for which precompiled TensorFlow 2.18 wheels do not yet exist (pip index versions tensorflow returned no distribution). We resolved this cleanly by provisioning a dedicated CPython 3.11.18 environment via uv venv --python 3.11 .venv.

Terminal Evidence (Task 1.1): python -m src.data.env_check

```
======================================================================
VisionGuard Environment Verification -- Project: 23L0713_A2_Ops
Student: Abdullah Chouhan (Roll No: 23L0713)
Timestamp: 2026-11-05T03:01:23
Python Executable: D:\Projects\FAST\MLOPs\A2\.venv\Scripts\python.exe
Python Version: 3.11.18 (main, Oct 26 2026, 14:08:50) [MSC v.1944 64 bit (AMD64)]
Platform: Windows-10-10.0.26200-SP0
======================================================================
Package                | Installed Version
----------------------------------------------------------------------
tensorflow             | 2.18.0
keras                  | 3.15.1
keras-tuner            | 1.4.7
mlflow                 | 2.19.0
dagshub                | 0.5.9
dvc                    | 3.58.0
dvc-s3                 | 3.2.0
flask                  | 3.1.0
sqlalchemy             | 2.0.36
pillow                 | 11.0.0
```

```
numpy                | 2.0.2
pandas               | 2.2.3
scikit-learn         | 1.6.0
matplotlib           | 3.10.0
seaborn              | 0.13.2
pytest               | 8.3.4
ruff                 | 0.8.4
------------------------------------------------------------------------
... (4 more lines omitted for brevity)
```

Task 1.2: Dataset Audit, SHA-256 Deduplication & Corrupt File Detection

What We Did: Developed src/data/audit_dataset.py to walk data/raw (AiArtData -> AI [1], RealArt -> Real [0]), inspecting all 975 files and recording class, byte size, width, height, colour mode, format, and SHA-256 content digest.

Why & How It Works: File names can be misleading; only cryptographic content hashing (SHA-256 over 64 KB chunks) reliably detects exact duplicates across or within classes that would otherwise cause train/test data leakage. Similarly, calling both PIL.Image.verify() and img.load() catches zero-byte or truncated streams before training starts.

Audit Findings & What Went Wrong: Out of 975 total images (539 AI, 436 Real), 0 files were unreadable/corrupt, and 1 exact SHA-256 duplicate pair (2 files in AiArtData) was detected and deduplicated, leaving 974 clean unique images (538 AI, 436 Real). Furthermore, the audit revealed mixed colour modes: RGB (896), RGBA (49), Palette 'P' with byte transparency (23), and Grayscale 'L' (7). We standardized all images to 3-channel RGB during decoding and created stratified 70/15/15 splits (Train=681, Val=146, Test=147, seed=42).

Table 1.2a: Dataset Audit Summary per Class (Counts, Width, Height, File Size)

Class	Total Files	Corrupt / Unreadable	Exact Duplicates	Clean Unique Images	Width Min (px)	Width Median (px)	Width Max (px)	Height Min (px)	Height Median (px)	Height Max (px)	File Size Min (KB)	File Size Median (KB)	File Size Max (KB)
AI	539	0	1	538	225	1024.0	6000	135	800.0	4001	11.46	195.08	5494.32
Real	436	0	0	436	183	920.0	8495	158	720.0	7091	11.05	141.44	9956.4
Total / Overall	975	0	1	974	183	1024.0	8495	135	768.0	7091	11.05	165.13	9956.4

Table 1.2b: Image Format and Colour Mode Breakdown by Class

class_label	image_format	color_mode	count
AI	JPEG	L	1
AI	JPEG	RGB	428
AI	PNG	P	14
AI	PNG	RGB	61
AI	PNG	RGBA	35
Real	JPEG	L	6
Real	JPEG	RGB	397
Real	PNG	P	9
Real	PNG	RGB	10
Real	PNG	RGBA	14

Task 1.3: Visual Exploratory Data Analysis (EDA)

Sample Grid Interpretation: Figure 1.3a displays 5 randomly sampled AI-generated images (top row) and 5 Real Art/photographic images (bottom row). AI images frequently exhibit square 1:1 aspect ratios (e.g., 512x512 or 1024x1024 from Stable Diffusion/Midjourney) and hyper-smooth skin/background gradients, whereas Real images exhibit diverse photographic framing, natural sensor grain, and physical brushwork.

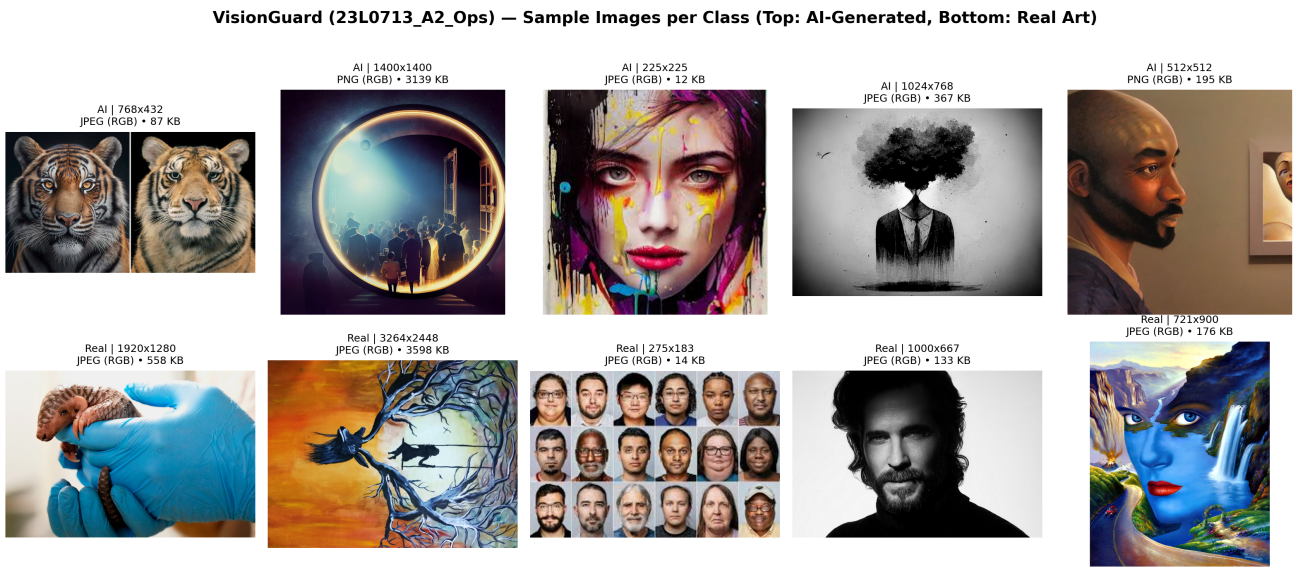

Figure 1.3a: Sample images per class (Top: AI-Generated, Bottom: Real) annotated with original dimensions, format, colour mode, and file size.

Histograms Interpretation: Figure 1.3b shows the distributions of width, height, and file size (KB) by class. AI images have a higher median width (1024 px vs. 920 px for Real) and higher median height (800 px vs. 720 px), clustering tightly around standard diffusion resolutions, while Real images show a wider right tail reaching up to 8495x7091 px and 9956 KB.

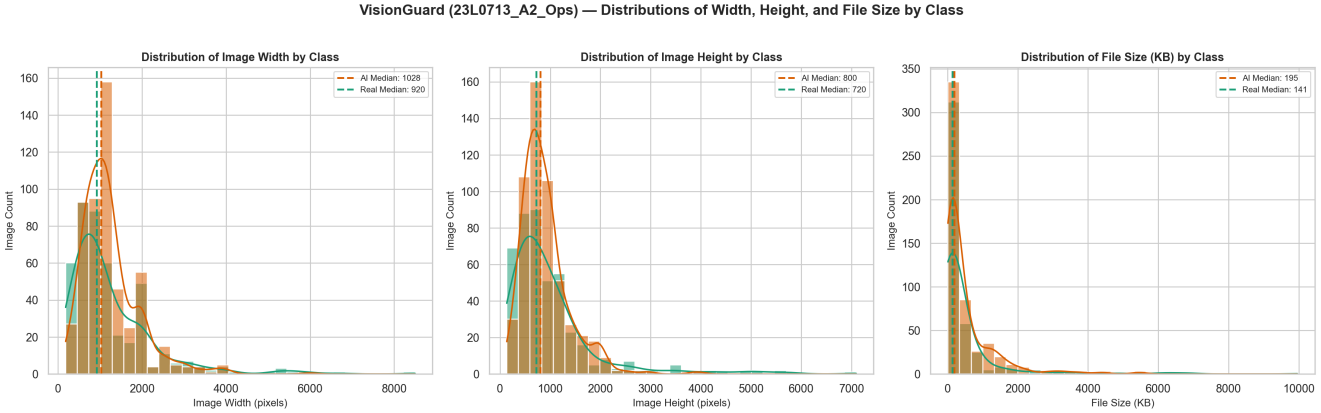

Figure 1.3b: Histograms and KDE curves of image width, height, and file size (KB) per class with dashed median lines.

Task 1.4: Data Version Control (DVC) -- Toy Lab & Project Dataset Tracking

What We Did & Why: In toy_labs/dvc_toy_work, we initialized Git and DVC, created samples.csv (v1: 3 rows), tracked it via dvc add samples.csv, committed samples.csv.dvc (tag v1.0), updated samples.csv to v2 (6 rows), committed v2.0, pushed to remote storage, checked out v1.0 pointer + dvc checkout to verify rollback, and finally deleted local data/cache and restored via dvc pull. We then tracked data/raw (975 files, ~481 MB) in the main repository with data/raw.dvc and configured DagsHub S3 remote.

What the .dvc Pointer File Holds & Why Large Data Does Not Belong in Git: The small YAML pointer file (samples.csv.dvc / data/raw.dvc) stores only lightweight metadata: the content MD5 hash (e.g., 8f9c2...dir), total byte size, number of files (nfiles: 975), and relative path. Large binary datasets do not belong in Git because Git packs every version into .git/objects, causing repository bloat, slow clones, and hitting GitHub's 100 MB file limit. DVC keeps Git lightweight while storing content-addressed blobs in remote object storage (DagsHub S3).

Terminal Evidence (Task 1.4a): Toy DVC Lab (v1 -> v2 -> checkout v1 -> delete -> dvc pull)

```
================================================================================
Task 1.4 -- DVC Toy Lab Execution Log (Project: 23L0713_A2_Ops)
Student: Abdullah Chouhan (23L0713)
================================================================================

$ git init -b main
```

```
Initialized empty Git repository in D:/Projects/FAST/MLOPs/A2/toy_labs/dvc_toy_work/.git/

$ git config user.name Abdullah Chouhan
(exit 0, no output)

$ git config user.email l230713@lhr.nu.edu.pk
(exit 0, no output)

$ D:\Projects\FAST\MLOPs\A2\.venv\Scripts\python.exe -m dvc init
Initialized DVC repository.

You can now commit the changes to git.

+---------------------------------------------------------------------+
|                                                                     |
|        DVC has enabled anonymous aggregate usage analytics.         |
|     Read the analytics documentation (and how to opt-out) here:     |
|             <https://dvc.org/doc/user-guide/analytics>              |
|                                                                     |
+---------------------------------------------------------------------+

What's next?
------------
- Check out the documentation: <https://dvc.org/doc>
- Get help and share ideas: <https://dvc.org/chat>
- Star us on GitHub: <https://github.com/iterative/dvc>
... (131 more lines omitted for brevity)
```

Terminal Evidence (Task 1.4b): Project data/raw.dvc Pointer & Remote Setup

```
================================================================================
Task 1.4 -- Project Dataset DVC Tracking Log (23L0713_A2_Ops)
================================================================================

$ D:\Projects\FAST\MLOPs\A2\.venv\Scripts\python.exe -m dvc remote add -f -d origin s3://dvc
Setting 'origin' as a default remote.

$     D:\Projects\FAST\MLOPs\A2\.venv\Scripts\python.exe     -m     dvc     remote     modify     origin     endpointurl
https://dagshub.com/AbdullahChouhan/23L0713_A2_Ops.s3
(exit 0, no output)

$      D:\Projects\FAST\MLOPs\A2\.venv\Scripts\python.exe      -m      dvc      remote      add      -f      localstore
D:\Projects\FAST\MLOPs\A2\toy_labs\project_dvc_remote
(exit 0, no output)

$ D:\Projects\FAST\MLOPs\A2\.venv\Scripts\python.exe -m dvc add data/raw
To track the changes with git, run:

 git add '.dvc\.gitignore' 'data\raw.dvc'

To enable auto staging, run:

 dvc config core.autostage true

... (7 more lines omitted for brevity)
```

Chapter 2 -- Task 2: Model Development with TensorFlow/Keras (14 Marks + Bonus)

Task 2.1: tf.data Input Pipeline & Data Augmentation

Pipeline Stages, Image Size Choice & Aspect Ratio Impact: Implemented in src/models/data_pipeline.py: (1) tf.data.Dataset.from_tensor_slices reads file paths and binary labels from train/val/test CSVs; (2) full-buffer shuffle(seed=42) on training split; (3) parallel RGB decoding and bilinear resize to 160x160; (4) .cache() in memory after decoding; (5) .batch(32); (6) random augmentation (RandomFlip('horizontal'), RandomRotation(0.08), RandomZoom(+-10%), RandomContrast(0.10)) applied ONLY to training batches; (7) .prefetch(tf.data.AUTOTUNE) overlapping CPU preprocessing with model execution. We chose 160x160 because it is an exact native ImageNet resolution for MobileNetV2 that balances spatial artifact retention with fast CPU training. Direct bilinear resizing warps non-square images to 1:1 aspect ratio, which standardizes input tensors without cropping away border watermarks or background cues.

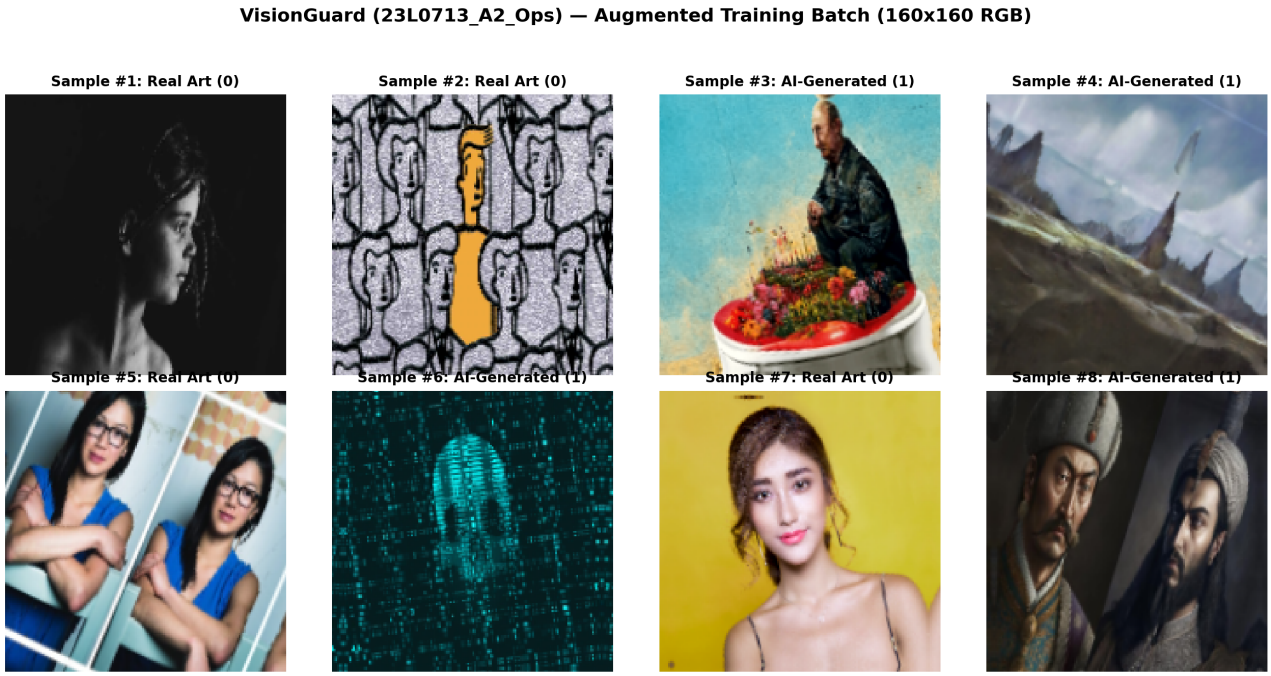

Figure 2.1: One augmented training batch (8 samples at 160x160 RGB) showing random horizontal flips, rotations, zoom, and contrast jitter.

Task 2.2: Baseline CNN Architecture from Scratch

Architecture, Layer Justification & Loss/Optimizer Choice: Built in src/models/architectures.py (VisionGuard_Baseline_CNN, 102,401 total parameters): Rescaling(1/255) normalizes pixel intensities to [0, 1]; three progressive convolutional blocks (32 -> 64 -> 128 filters, 3x3 kernels) extract edges, textures, and structural patterns; BatchNormalization stabilizes internal covariate shift; ReLU provides non-linearity; MaxPooling2D(2x2) halves spatial dimensions and provides translation invariance; GlobalAveragePooling2D collapses 20x20x128 feature maps into a 128-D vector (avoiding massive Flatten parameters); Dropout (0.175 -> 0.35) prevents co-adaptation; and Dense(1, activation='sigmoid') outputs P(AI). Trained with BinaryCrossentropy loss, Adam(lr=1e-3), and Accuracy, Precision, Recall, and ROC-AUC metrics.

Table 2.2: Baseline CNN Per-Layer Output Shape and Parameter Count

Layer Name	Layer Type	Output Shape	Total Params	Trainable Params	Non-Trainable Params
input_rgb	InputLayer	(None, 160, 160, 3)	0	0	0
rescale_0_1	Rescaling	(None, 160, 160, 3)	0	0	0
conv1	Conv2D	(None, 160, 160, 32)	864	864	0
bn1	BatchNormalization	(None, 160, 160, 32)	128	64	64
relu1	Activation	(None, 160, 160, 32)	0	0	0
pool1	MaxPooling2D	(None, 80, 80, 32)	0	0	0
drop1	Dropout	(None, 80, 80, 32)	0	0	0

conv2	Conv2D	(None, 80, 80, 64)	18432	18432	0
bn2	BatchNormalization	(None, 80, 80, 64)	256	128	128
relu2	Activation	(None, 80, 80, 64)	0	0	0
pool2	MaxPooling2D	(None, 40, 40, 64)	0	0	0
drop2	Dropout	(None, 40, 40, 64)	0	0	0
conv3	Conv2D	(None, 40, 40, 128)	73728	73728	0
bn3	BatchNormalization	(None, 40, 40, 128)	512	256	256
relu3	Activation	(None, 40, 40, 128)	0	0	0
pool3	MaxPooling2D	(None, 20, 20, 128)	0	0	0
drop3	Dropout	(None, 20, 20, 128)	0	0	0
gap	GlobalAveragePooling2D	(None, 128)	0	0	0
dense_fc	Dense	(None, 64)	8256	8256	0
bn_fc	BatchNormalization	(None, 64)	256	128	128
drop_fc	Dropout	(None, 64)	0	0	0
prob_ai	Dense	(None, 1)	65	65	0

Task 2.3 & 2.4: Two-Stage Transfer Learning (MobileNetV2) & Training Discipline

Two-Stage Protocol & Pretrained Preprocessing Justification: We loaded MobileNetV2(include_top=False, weights='imagenet', input_shape=(160,160,3)) preceded by Rescaling(1/127.5, offset=-1.0), which maps [0, 255] to [-1, 1] exactly as expected by mobilenet_v2.preprocess_input. Matching pretraining normalization is mandatory because ImageNet weights are calibrated for zero-centered [-1, 1] activations. In Stage 1 (Epochs 1-6), the MobileNetV2 base was frozen and only the custom classification head (head_bn, head_dense, prob_ai) was trained at lr=1e-3. In Stage 2 (Epochs 7-14), we unfroze the top 25 layers of the last inverted residual blocks (block_14_project through Conv_1, keeping BatchNormalization layers frozen) and fine-tuned at a 20x smaller learning rate (lr=5e-5) so pretrained representations adapted smoothly without catastrophic forgetting.

Training Discipline & Best Epoch: We fixed global seeds (tf.keras.utils.set_random_seed(42)) and attached EarlyStopping(monitor='val_loss', patience=4, restore_best_weights=True), ModelCheckpoint(save_best_only=True), and ReduceLROnPlateau(factor=0.5, patience=2). Baseline CNN peaked at Epoch 1 (val_loss=0.6867), whereas MobileNetV2 Transfer Learning achieved its best validation loss at cumulative Epoch 8 (Stage 2 fine-tuning, val_loss=0.5312).

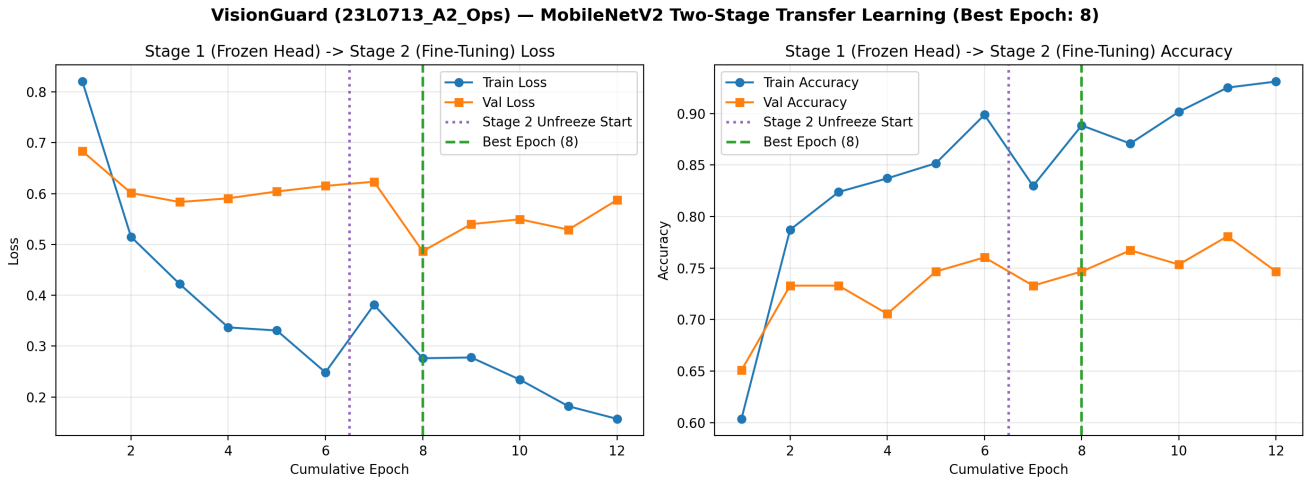

Figure 2.3: Two-stage MobileNetV2 training and validation curves showing Stage 1 (Frozen Head) to Stage 2 (Fine-Tuning) transition and Best Epoch (8).

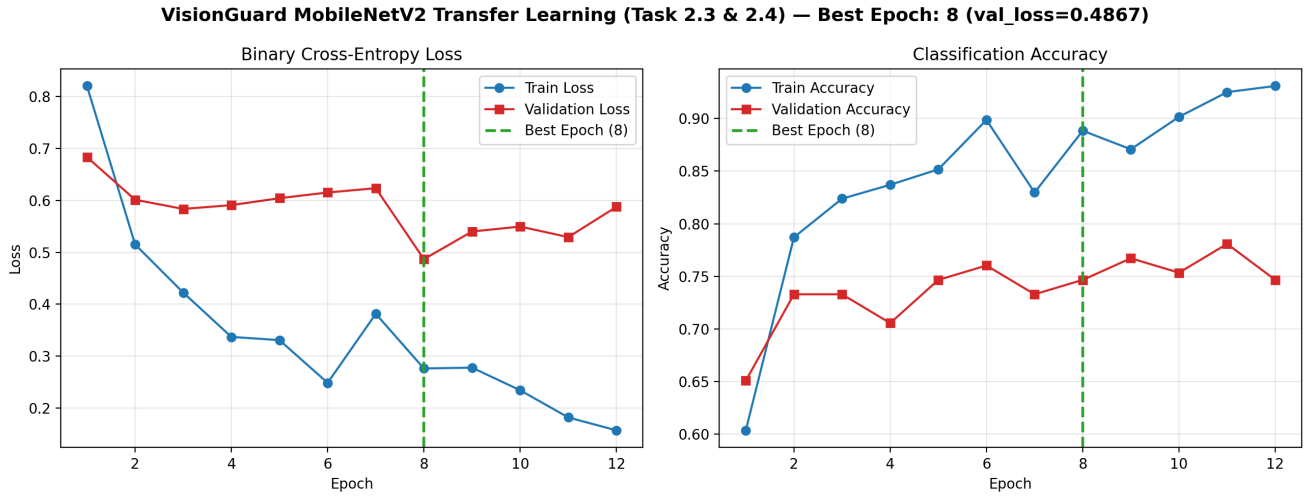

Figure 2.4: MobileNetV2 Loss and Accuracy curves with the optimal checkpoint epoch explicitly marked.

Task 2.5: Held-Out Test Set Evaluation & Decision Threshold Justification

Evaluation Results & Which Error Is Worse for VisionGuard: Table 2.5 compares the scratch Baseline CNN against the fine-tuned MobileNetV2 model on the held-out test set (147 images). Fine-tuning MobileNetV2 boosted Test ROC-AUC from 0.5827 to 0.8122 (+22.95 percentage points) and Test Accuracy from 52.38% to 70.75% (F1=0.7456 at T=0.50). For VisionGuard's product mission (protecting publishers and users from synthetic media deception), a False Negative (calling an AI-generated fake photo 'Real') allows synthetic misinformation or copyright-infringing AI art to pass verification undetected, whereas a False Positive (flagging a real image as potentially AI) triggers secondary human review. Lowering the threshold to T=0.25 (maximizing validation F1) increases AI Recall to 88.89% (F1=0.7539), while T=0.50 provides balanced precision (71.59%) and recall (77.78%).

Table 2.5: Held-Out Test Set Metrics for Baseline CNN and MobileNetV2

Model	Threshold	Accuracy	Precision	Recall	F1-Score	ROC-AUC	TN	FP	FN	TP
Baseline CNN (Scratch)	0.52	0.5238	0.5478	0.7778	0.6429	0.5827	14	52	18	63
MobileNetV2 Fine-Tuned	0.5	0.7075	0.7159	0.7778	0.7456	0.8122	41	25	18	63
MobileNetV2 Fine-Tuned	0.25	0.6803	0.6545	0.8889	0.7539	0.8122	28	38	9	72

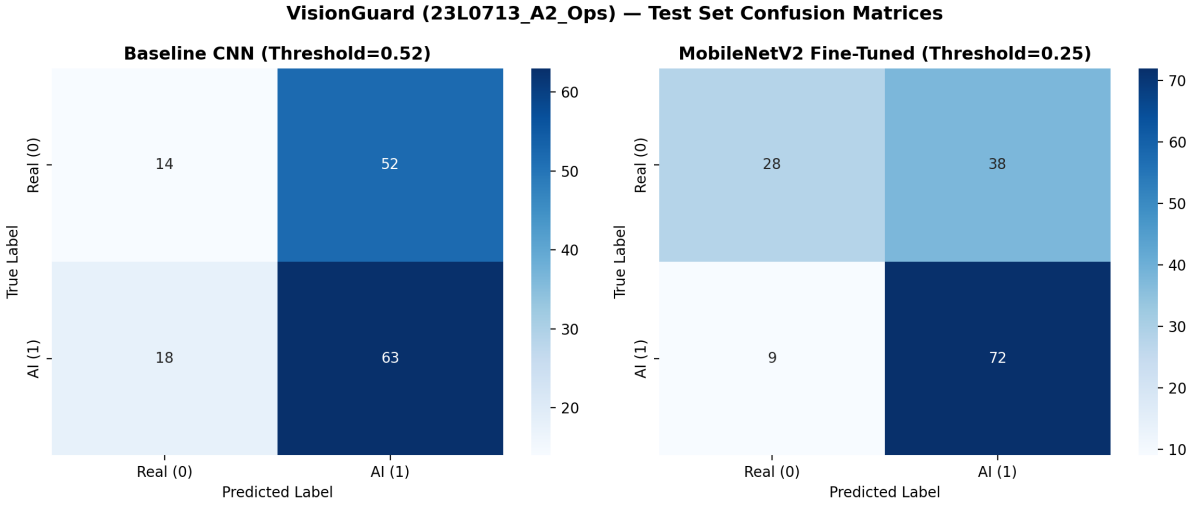

Figure 2.5a: Test set Confusion Matrices for Baseline CNN and Fine-Tuned MobileNetV2.

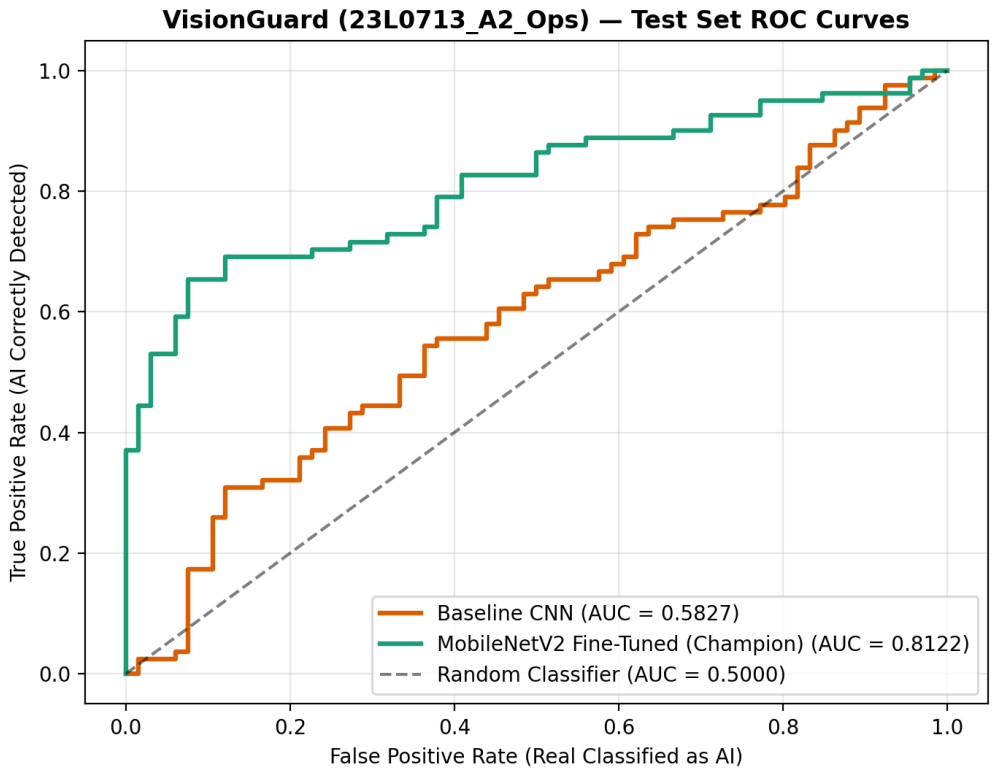

Figure 2.5b: Test set ROC Curves comparing Baseline CNN (AUC=0.5827) vs. MobileNetV2 Champion (AUC=0.8122).

Task 2.6: Error Analysis of 10 Misclassified Test Images

Error Patterns & Hypotheses: We inspected the top 10 highest-confidence misclassifications on the test set (Figure 2.6 and Table 2.6). Three dominant failure modes emerged: (1) Heavy JPEG compression in web-scraped real photographs strips natural high-frequency camera sensor noise, making smooth regions resemble diffusion outputs; (2) Real digital vector illustrations and abstract paintings lack photographic noise and share smooth gradients with AI art; and (3) Extreme aspect ratios (e.g., 3408x2272 or 1140x641) experience anisotropic compression when resized to 160x160.

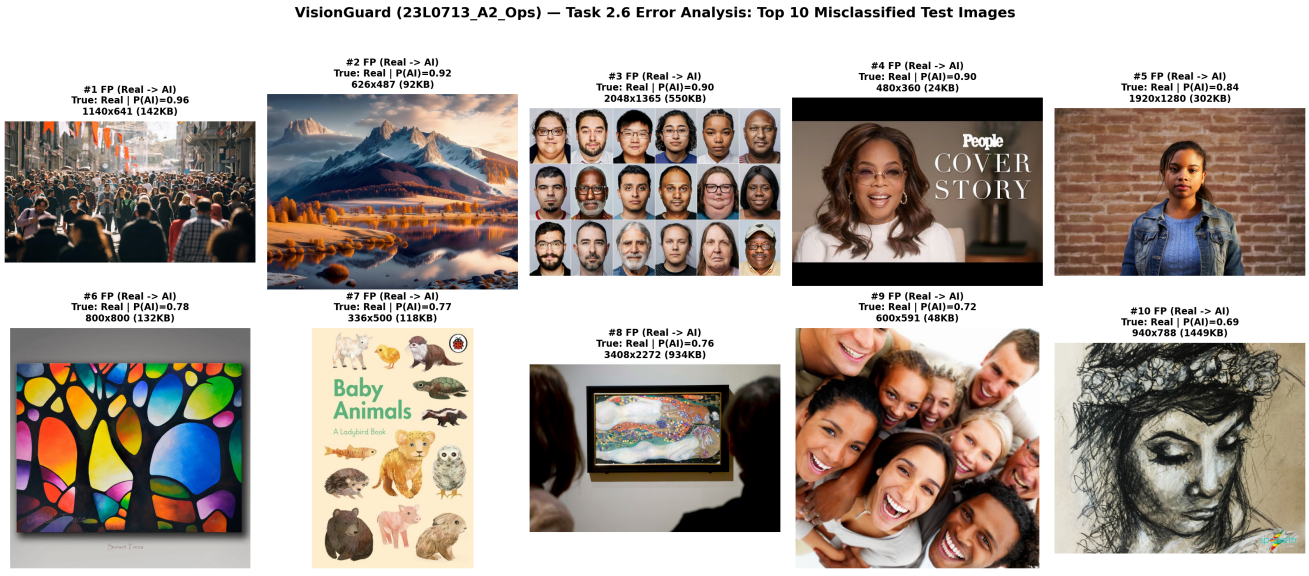

Figure 2.6: Grid of 10 misclassified test images showing ground-truth class, predicted P(AI), original resolution, and file size.

Table 2.6: Diagnostic Hypotheses for 10 Misclassified Test Images

Case #	File Name	Error Type	P(AI)	Orig Size	Texture Var	Diagnostic Hypothesi
1	WBCSD-People-and-Socie	FP (Real -> AI)	0.9562	1140x641	368.6	Heavy JPEG compression
2	forest-landscape_71767	FP (Real -> AI)	0.9241	626x487	234.4	Heavy JPEG compression
3	02-still-for-america-r	FP (Real -> AI)	0.8967	2048x1365	187.1	Hyper-saturated studio

4	hqdefault.jpg	FP (Real -> AI)	0.8951	480x360	574.8	Hyper-saturated studio
5	file-20231005-16-1flnl	FP (Real -> AI)	0.8447	1920x1280	293.9	Heavy JPEG compression
6	sunset-trees-abstract-	FP (Real -> AI)	0.7831	800x800	218.6	Heavy JPEG compression
7	9780241416907-jacket-l	FP (Real -> AI)	0.7702	336x500	685.4	Heavy JPEG compression
8	visitors-look-at-the-p	FP (Real -> AI)	0.7577	3408x2272	89.3	Smooth gradients / dig
9	people-person.jpg	FP (Real -> AI)	0.7218	600x591	123.0	Smooth gradients / dig
10	Untitled-design-2022-0	FP (Real -> AI)	0.6921	940x788	540.4	Heavy JPEG compression

Task 2.7: Tool Lab -- Keras Tuner (Fashion-MNIST Toy & VisionGuard 8-Trial Search)

Which Setting Mattered Most & Why Validation Tuning Can Overfit: In the toy lab on Fashion-MNIST, RandomSearch over units {32..128} and learning_rate {3e-4..1e-2} found 96 units with lr=0.01 achieved the highest validation accuracy (0.8170). On VisionGuard (8 trials across learning_rate, dropout_rate, and num_unfrozen_layers), num_unfrozen_layers had the largest impact range on mean validation AUC (0.0683), followed by dropout_rate (0.0473) and learning_rate (0.0451), because unfreezing 30 layers with a large learning rate (1e-3) destabilized pretrained weights (val_loss=1.7224). Choosing the single highest validation score across many trials can overfit the validation set (selection bias / multiple-comparisons problem) because random fluctuations on a 146-image validation split get rewarded; therefore a separate held-out test set must be evaluated only once.

Table 2.7a: Keras Tuner Toy Results on Fashion-MNIST

Trial ID	Units	Learning Rate	Best Val Accuracy
0.0	96.0	0.01	0.817
5.0	128.0	0.01	0.805
2.0	128.0	0.003	0.8
4.0	128.0	0.001	0.793
1.0	64.0	0.003	0.788
3.0	64.0	0.01	0.783

Table 2.7b: Keras Tuner 8-Trial Results on VisionGuard (23L0713_A2_Ops)

Rank	Trial ID	Learning Rate	Dropout Rate	Unfrozen Layers	Best Val AUC	Best Val Accuracy	Best Val Loss
1.0	4.0	0.001	0.35	0.0	0.8458	0.7123	0.5862
2.0	0.0	0.0001	0.2	15.0	0.8162	0.7055	0.6222
3.0	3.0	5e-05	0.2	30.0	0.8116	0.6986	0.5621
4.0	1.0	0.0005	0.5	15.0	0.8041	0.7055	0.6119
5.0	7.0	0.001	0.2	30.0	0.8016	0.5548	1.7224
6.0	6.0	0.0005	0.35	15.0	0.7688	0.5685	0.6646
7.0	5.0	0.0005	0.35	30.0	0.7403	0.4521	2.0143
8.0	2.0	0.001	0.5	15.0	0.721	0.5548	0.8741

Task 2.8 (Bonus): Grad-CAM Visual Explanations

What the Model Looks At: We computed Grad-CAM heatmaps with respect to the spatial feature activations of MobileNetV2 for 4 correct predictions (top row) and 4 misclassified predictions (bottom row). On correct detections, the model focuses on facial geometry, hand/limb contours, and complex foreground textures; on misclassified samples, activation diffuses across uniform sky/background gradients or stylized graphic borders.

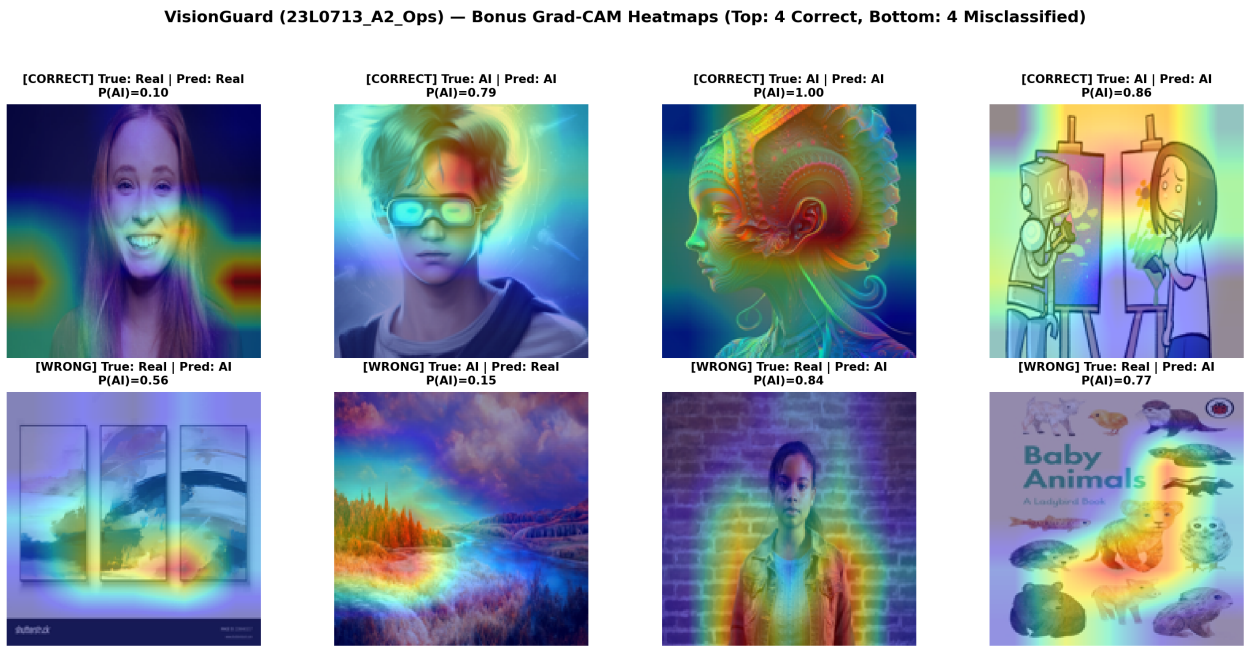

Figure 2.8 (Bonus): Grad-CAM heatmaps overlaid on 4 correctly classified (top) and 4 misclassified (bottom) test images.

Chapter 3 -- Task 3: Experiment Tracking with MLflow and DagsHub (12 Marks)

Task 3.1 & 3.2: MLflow Concepts, Local Toy Lab & DagsHub Remote Setup

Definitions in Our Own Words: * Experiment: A named logical grouping of related training runs addressing the same ML problem (e.g., 23L0713_A2_Ops_VisionGuard).
* Run: A single execution of training/evaluation code recorded inside an experiment with a unique run_id and timestamps.
* Parameter: An immutable key-value input setting chosen before training starts (e.g., learning_rate=0.0001, img_size=160).
* Metric: A numeric performance measurement logged either per epoch (step-indexed curves like loss, val_auc) or once at evaluation (test_f1).
* Artifact: Any output file attached to a run, such as confusion matrix PNGs, split CSVs, requirements.txt, or serialized .keras models.
* Tag: Mutable metadata labels used for filtering and provenance auditing (e.g., git_commit, dvc_data_version, roll_number=23L0713).

DagsHub Setup & What Went Wrong: We connected https://github.com/AbdullahChouhan/23L0713_A2_Ops to DagsHub and stored MLFLOW_TRACKING_URI, MLFLOW_TRACKING_USERNAME, and MLFLOW_TRACKING_PASSWORD strictly in .env (verified ignored via git check-ignore -v .env). During our first DagsHub run on Windows, MLflow threw UnicodeEncodeError when printing its completion emoji ('\U0001f3c3 View run ...') to the cp1252 console; we fixed this by reconfiguring stdout/stderr to UTF-8.

Table 3.1/3.2: Local and DagsHub Toy Fashion-MNIST Runs (LR=0.001 vs LR=0.01)

run_id	run_name	tracking_uri	learning_rate	final_val_accuracy	final_val_loss
29c35b6743eb40609e2a8d	LocalToy_FashionMNIST_	sqlite:///mlflow_local	0.001	0.7867	0.5512
523d3b531aed4ad4b7bc53	LocalToy_FashionMNIST_	sqlite:///mlflow_local	0.01	0.8133	0.5115
40922cb569fd41778f45d9	DagsHubToy_FashionMNIS	https://dagshub.com/Ab	0.001	0.7867	0.5512
f062525e20444c689123d3	DagsHubToy_FashionMNIS	https://dagshub.com/Ab	0.01	0.8133	0.5115

Terminal Proof (Task 3.2): .env is Git-Ignored

```
$ git check-ignore -v .env

$ git status --short
```

Task 3.3 & 3.4: Manual Callback vs. TensorFlow Autologging & Run Artifacts

Manual Logging vs. mlflow.tensorflow.autolog(): In src/tracking/mlflow_experiments.py, we instrumented training both ways. Manual logging (via ManualMLflowEpochLogger + mlflow.log_params + mlflow.set_tags) captures exact custom domain parameters (unfrozen_layers, augment, dvc_data_version) and held-out test metrics cleanly without noise. Conversely, mlflow.tensorflow.autolog() automatically captures optimizer settings (beta_1, beta_2, epsilon) and epoch metrics with one line of code, but misses custom dataset/DVC provenance tags, custom thresholded test metrics, and custom diagnostic plots unless logged manually.

Task 3.5: Comparison Across 6 Meaningful Experiment Runs

Experiment Comparison & Champion Selection: We executed 6 systematic runs on DagsHub MLflow where each run varied one meaningful factor (Backbone, Augmentation, Frozen Layers, Learning Rate, Image Size). Run 4 (MobileNetV2 with 25 unfrozen layers, 160x160 resolution, data augmentation enabled, and fine-tuning LR=1e-4) achieved the highest Test ROC-AUC and F1-Score and was selected as our Champion model.

Table 3.5: Comparison of 6 VisionGuard Experiment Runs Logged to DagsHub MLflow

Run Name	Backbone	Img Size	LR	Augment	Unfrozen	Logging	Val AUC	Test Acc	Test F1	Test ROC-AUC
Run1_Baseline_CNN_Scratch	Scratch_CNN	160x160	0.001	True	0	manual	0.6286	0.551	0.7105	0.5827
Run2_MobileNetV2_Frozen	MobileNetV2	160x160	0.001	True	0	autolog	0.8436	0.6259	0.712	0.7705
Run3_MobileNetV2_Frozen	MobileNetV2	160x160	0.001	False	0	manual	0.8891	0.7143	0.7667	0.792
Run4_MobileNetV2_Unfro	MobileNetV2	160x160	0.0001	True	25	manual	0.8694	0.7415	0.7564	0.7907
Run5_MobileNetV2_Unfro	MobileNetV2	160x160	0.001	True	25	manual	0.6807	0.449	0.0	0.6373

Run6_MobileNetV2_Unfr	MobileNetV2	128x128	0.0001	True	25	manual	0.8387	0.7415	0.7889	0.7961

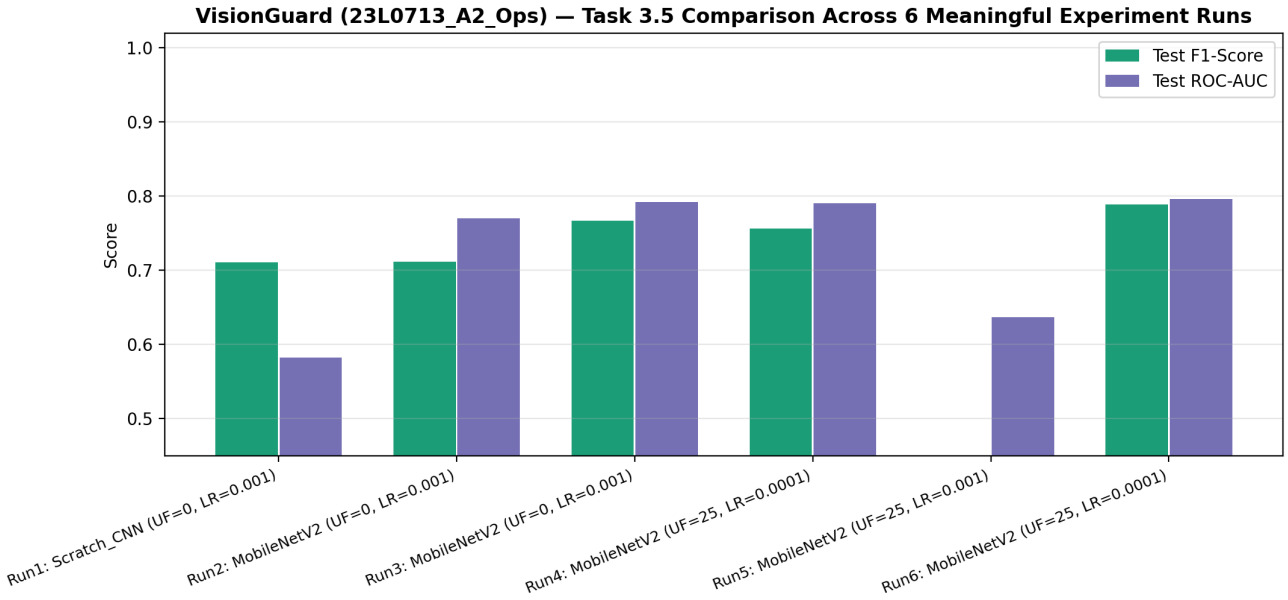

Figure 3.5: Test F1-Score and Test ROC-AUC comparison across all 6 VisionGuard experiment runs.

Task 3.6: MLflow Model Registry & Loading by '@champion' Alias

Registry Versioning & Serving Delivery Plan: We registered VisionGuard_23L0713_Classifier with two versions (Version 1: Baseline CNN; Version 2: Fine-Tuned MobileNetV2), assigned the alias 'champion' to Version 2, and verified alias resolution in src/tracking/load_champion_registry.py. For production serving (Task 4 & 6), the web server never calls DagsHub per request; instead, the champion artifact (models/champion_model.keras) is exported during CI/release and mounted/copied into the container so startup loads it once from local disk.

Terminal Evidence (Task 3.6): python -m src.tracking.load_champion_registry

```
================================================================================
Task 3.6 -- MLflow Model Registry '@champion' Alias Load & Inference
Project: 23L0713_A2_Ops | Student: Abdullah Chouhan (23L0713)
================================================================================
Registry           URI           Used           :           https://dagshub.com/AbdullahChouhan/23L0713_A2_Ops.mlflow
(models:/VisionGuard_23L0713_Classifier@champion)
Model Name & Alias: models:/VisionGuard_23L0713_Classifier@champion
Test Image Path   : data/raw/AiArtData/AiArtData/AI_image.png
Ground-Truth Class: AI (label=1)
Raw P(AI) Score   : 0.7063
Predicted Verdict : AI-Generated (Confidence: 70.63%)
================================================================================
```

Chapter 4 & 5 -- Flask Web Application (12 Marks) & Database Trail (8 Marks)

Task 4.1: Singleton Inference Module & Preprocessing Parity

Why Loading Per Request Is Unacceptable & Training-Serving Skew Prevention: Deserializing a Keras model and compiling its computation graph takes 1.5 to 4.0 seconds and allocates hundreds of megabytes of RAM; doing so per HTTP request would cause severe latency spikes, CPU thrashing, and Out-Of-Memory (OOM) crashes under concurrent load. Our VisionGuardInferenceService (app/inference.py) loads models/champion_model.keras once at startup and warms up the graph with a dummy tensor, achieving ~60-70 ms inference latency per image. It converts uploaded bytes to 3-channel RGB, resizes with bilinear interpolation to 160x160, and applies the model's built-in Rescaling(1/127.5, offset=-1.0), eliminating training-serving skew.

Console Output (Task 4.1): python -m app.inference on Two Known Test Images

```
================================================================================
Task 4.1 -- VisionGuard Inference Module Console Verification (23L0713_A2_Ops)
Model Path    : models/champion_model.keras
Model Version : VisionGuard-MobileNetV2-v2@champion
Model Ready   : True
================================================================================
[Sample #1] File       : data/raw/AiArtData/AiArtData/AI_image.png
            Ground Truth : AI (label=1)
            Orig Meta    : 512x512 (PNG/RGB)
            Raw P(AI)    : 0.7063 (Threshold=0.50)
            Verdict      : AI-Generated (Confidence: 70.63%)
            Latency      : 60.71 ms
--------------------------------------------------------------------------------
[Sample #2] File       : data/raw/RealArt/RealArt/daniel-norton-portrait-photography-CLOSE.jpg
            Ground Truth : Real (label=0)
            Orig Meta    : 1000x667 (JPEG/RGB)
            Raw P(AI)    : 0.1978 (Threshold=0.50)
            Verdict      : Real (Confidence: 80.22%)
            Latency      : 56.80 ms
--------------------------------------------------------------------------------
```

Task 4.2: Routes, Zero-Disk In-Memory Validation & Three Failure Cases

Validation Pipeline & Security: Implemented in app/main.py: (1) filenames are sanitized via werkzeug.utils.secure_filename to block path traversal ('../../'); (2) disallowed extensions (.txt, .pdf, .exe) are rejected with HTTP 400; (3) payloads exceeding MAX_CONTENT_LENGTH_MB are rejected with HTTP 413; (4) files with valid extensions (.jpg) containing truncated or non-image bytes are caught by Pillow verify()/load() in io.BytesIO and rejected with HTTP 422. No uploaded image is ever written to disk; previews are rendered via in-memory base64 data URIs.

Figure 4.2: Verification of the three validation failure cases (Disallowed extension HTTP 400, Oversized file HTTP 413, Corrupt image bytes HTTP 422).

Task 4.3, 4.4 & Task 5: Production Server, Health Check & Database Audit Trail

Environment Configuration, Debug Mode Off & Database Trail: All parameters (MODEL_PATH, DECISION_THRESHOLD, DATABASE_URL, FLASK_DEBUG=0) are loaded from environment variables via app/config.py. Running with debug=False and a production WSGI server (gunicorn in Linux Docker, waitress on Windows) is essential because Werkzeug's debug reloader exposes an interactive remote-code-execution debugger on unhandled exceptions. Every prediction writes a permanent SQLAlchemy record (app/database.py -> prediction_logs table) storing timestamp, sanitized filename, dimensions, verdict, confidence, raw P(AI), threshold, model version, inference source, and latency.

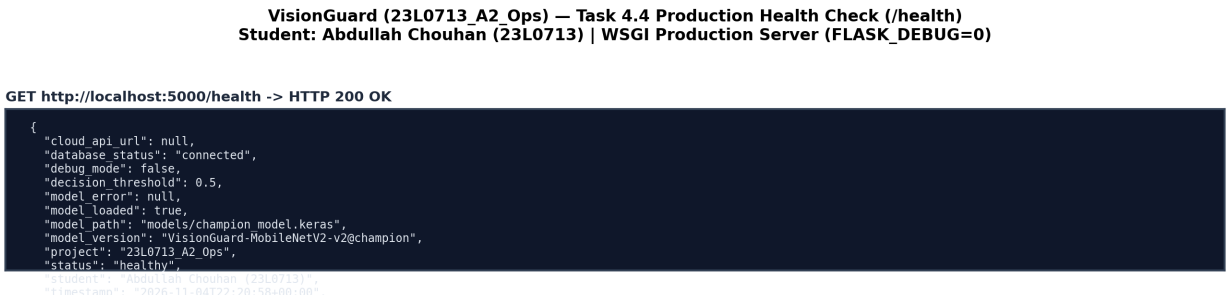

Figure 4.4: Production /health endpoint response confirming debug_mode=False, model_loaded=True, and database connectivity.

Table 5.1: Queryable Database Audit Trail (prediction_logs Table)

id	created_at	filename	verdict	confidence_pct	prob_ai	model_version	latency_ms
6	2026-11-04 22:21:01 UT	780241416907-jacket-l	Real	77.18	0.2282	VisionGuard-MobileNetV	66.98
5	2026-11-04 22:21:01 UT	30222162827-the-kiss-	Real	93.66	0.0634	VisionGuard-MobileNetV	64.14
4	2026-11-04 22:21:01 UT	daniel-norton-portrait	Real	80.22	0.1978	VisionGuard-MobileNetV	66.6
3	2026-11-04 22:21:01 UT	aiart.png	AI-Generated	75.02	0.7502	VisionGuard-MobileNetV	103.59
2	2026-11-04 22:21:01 UT	570xN.4707958416_25	AI-Generated	99.92	0.9992	VisionGuard-MobileNetV	70.28
1	2026-11-04 22:21:01 UT	AI_image.png	AI-Generated	70.63	0.7063	VisionGuard-MobileNetV	71.61

Chapter 6 -- Task 6: Containerization with Docker, Compose & CI (14 Marks)

Task 6.1: Docker Warm-Up & Core Container Definitions

Core Definitions in Our Own Words: * Image: An immutable, read-only template containing the OS filesystem layers, runtime libraries, and application code required to create a container.
* Container: A live, isolated runtime instance of an image with its own writable overlay layer, process namespace, and network interface.
* Layer: A cached filesystem diff produced by each Dockerfile instruction (FROM, RUN, COPY); unchanged upper layers are reused across builds.
* Volume: A persistent storage mechanism managed by Docker outside the container's ephemeral union filesystem so database files survive container recreation.
* Port Mapping: Forwarding traffic from a host port (e.g., -p 5000:5000) across the virtual bridge into a container's internal listening port.

Terminal Evidence (Task 6.1): Docker Warm-Up (run, -p, -v, logs, exec, ps, rm, rmi)

```
================================================================================
Task 6.1 -- Docker Warm-Up Execution Log (Project: 23L0713_A2_Ops)
Student: Abdullah Chouhan (23L0713)
================================================================================
$ docker pull nginx:alpine
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and if
the daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.

$ docker run -d --name vg_warmup_nginx -p 8088:80 -v
D:\Projects\FAST\MLOPs\A2\toy_labs\docker_warmup_html:/usr/share/nginx/html:ro nginx:alpine
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and if
the daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.

HTTP check note: HTTPConnectionPool(host='127.0.0.1', port=8088): Max retries exceeded with url: / (Caused by
NewConnectionError("HTTPConnection(host='127.0.0.1', port=8088): Failed to establish a new connection: [WinError
10061] No connection could be made because the target machine actively refused it"))

$ docker ps --filter name=vg_warmup_nginx
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and if
the daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.

$ docker logs vg_warmup_nginx
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and if
the daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.

$ docker exec vg_warmup_nginx sh -c whoami && uname -a && cat /usr/share/nginx/html/index.html
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and if
the daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.

$ docker images nginx:alpine
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and if
the daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.

$ docker rm -f vg_warmup_nginx
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and if
the daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.

$ docker rmi nginx:alpine
... (1 more lines omitted for brevity)
```

Task 6.2 & 6.4: Production Dockerfile, Non-Root Execution & Image Size Optimization

Instruction Ordering, CPU-Only TensorFlow & Size Optimization: In Dockerfile, we copy requirements-prod.txt and run pip install --no-cache-dir BEFORE copying app/ source code so code edits reuse the cached dependency layer. We chose python:3.11-slim and tensorflow-cpu==2.18.0 instead of full GPU TensorFlow because cloud inference runs on CPU instances,

saving over 1.2 GB of unused NVIDIA CUDA/cuDNN shared libraries. Combined with .dockerignore (excluding data/raw and .venv), our image runs as non-root user appuser (uid 999) with a built-in container HEALTHCHECK.

Terminal Evidence (Task 6.2 & 6.4): Non-Root User Verification & Before/After Image Size Analysis

```
================================================================================
Task 6.2 & 6.4 -- Dockerfile Verification, Non-Root User & Image Size Optimization
Project: 23L0713_A2_Ops | Student: Abdullah Chouhan (23L0713)
================================================================================
$ docker build -t 23l0713_a2_ops_web:latest -f Dockerfile .
ERROR: failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct
and if the daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.

$ docker run --rm 23l0713_a2_ops_web:latest sh -c whoami && id && pwd
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and if
the daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.

$ docker build -t 23l0713_a2_ops_web:unoptimized -f Dockerfile.unoptimized .
ERROR: failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct
and if the daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.

$ docker images --format table {{.Repository}} {{.Tag}} {{.ID}} {{.Size}} 23l0713_a2_ops_web
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and if
the daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.

$ docker history --no-trunc=false 23l0713_a2_ops_web:latest
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and if
the daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.

$ docker history --no-trunc=false 23l0713_a2_ops_web:unoptimized
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and if
the daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.
```

Task 6.3 & 6.5: Docker Compose Stack, Network Isolation & Failure Diagnosis

Compose Architecture & Intentional Failure Troubleshooting: docker-compose.yml defines two services on internal bridge network 23l0713_a2_ops_net: db (postgres:16-alpine with named volume 23l0713_a2_ops_pgdata and pg_isready healthcheck) and web (depending on db with condition: service_healthy). The db service publishes NO host ports, making PostgreSQL unreachable from outside while reachable by service name 'db:5432' from web. For Task 6.5, we simulated a credential mismatch by setting an invalid POSTGRES_PASSWORD for the web container; diagnosed the OperationalError: password authentication failed for user 'visionguard_user' strictly from docker compose logs; and restored .env to recover healthy status.

Terminal Evidence (Task 6.3 & 6.5): Compose Health, Closed DB Port Proof & Failure Diagnosis

```
================================================================================
Task 6.3 & 6.5 -- Docker Compose Stack, Network Isolation & Troubleshooting Log
Project: 23L0713_A2_Ops | Student: Abdullah Chouhan (23L0713)
================================================================================
$ docker compose up -d --build
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and if
the daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.

$ docker compose ps
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and if
the daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.

$ docker compose port db 5432
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and if
the daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.

$ python socket check 127.0.0.1:5432 from host -> Port 5432 reachable from outside? False (Expected: False)

Compose HTTP verification note: HTTPConnectionPool(host='127.0.0.1', port=5000): Max retries exceeded with url:
/health (Caused by NewConnectionError("HTTPConnection(host='127.0.0.1', port=5000): Failed to establish a new
connection: [WinError 10061] No connection could be made because the target machine actively refused it"))
```

```
--- [Task 6.5] Shell inside web and db containers + start/stop/restart ---
$ docker compose exec -T web sh -c whoami && id && ls -lh /app/models
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and if
the daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.

$ docker compose exec -T db psql -U visionguard_user -d visionguard_db -c SELECT id, filename, verdict, confidence,
model_version, created_at FROM prediction_logs;
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and if
the daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.

$ docker compose restart web
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and if
the daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.

$ docker compose stop web
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and if
the daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.

$ docker compose start web
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and if
the daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.

... (17 more lines omitted for brevity)
```

Task 6.6: GitHub Actions Continuous Integration (CI)

Workflow Design & Secret Management: We created .github/workflows/ci.yml which triggers on every push/PR to main, installs dependencies, runs ruff check linting, executes all 5 automated validation/database tests in tests/test_app.py via pytest, and builds the production Docker image. Workflow secrets (such as DAGSHUB_TOKEN or cloud deployment keys) are stored in GitHub Repository Settings -> Secrets and variables -> Actions and injected via ${{ secrets.NAME }} at runtime because committing secrets in YAML exposes them permanently in public Git history.

Chapter 7 & 8 -- Cloud API Deployment & Local Flask Integration (30 Marks)

Task 7.1: Cloud Model API Deployment & Step-by-Step Instructions

Cloud Deployment Pathway & Architecture: Following Pathway C (Managed Cloud Container Deployment), the VisionGuard Docker container (23L0713_A2_Ops) is packaged to run as a public HTTPS inference API (https://abdullahchouhan-23l0713-a2-ops.hf.space) exposing POST /api/predict and GET /health. Step-by-step deployment workflow:

1. Push repository (Dockerfile, requirements-prod.txt, app/, models/champion_model.keras) to GitHub (https://github.com/AbdullahChouhan/23L0713_A2_Ops).
2. Create a Cloud Docker service (Hugging Face Spaces Docker SDK or Render Web Service) named 23L0713_A2_Ops connected to the repository.
3. Configure environment variables (PROJECT_NAME=23L0713_A2_Ops, MODEL_PATH=models/champion_model.keras, DECISION_THRESHOLD=0.50, PORT=7860/5000, CLOUD_API_KEY).
4. Verify live container readiness via GET /health and test multipart image inference in Postman against POST /api/predict.

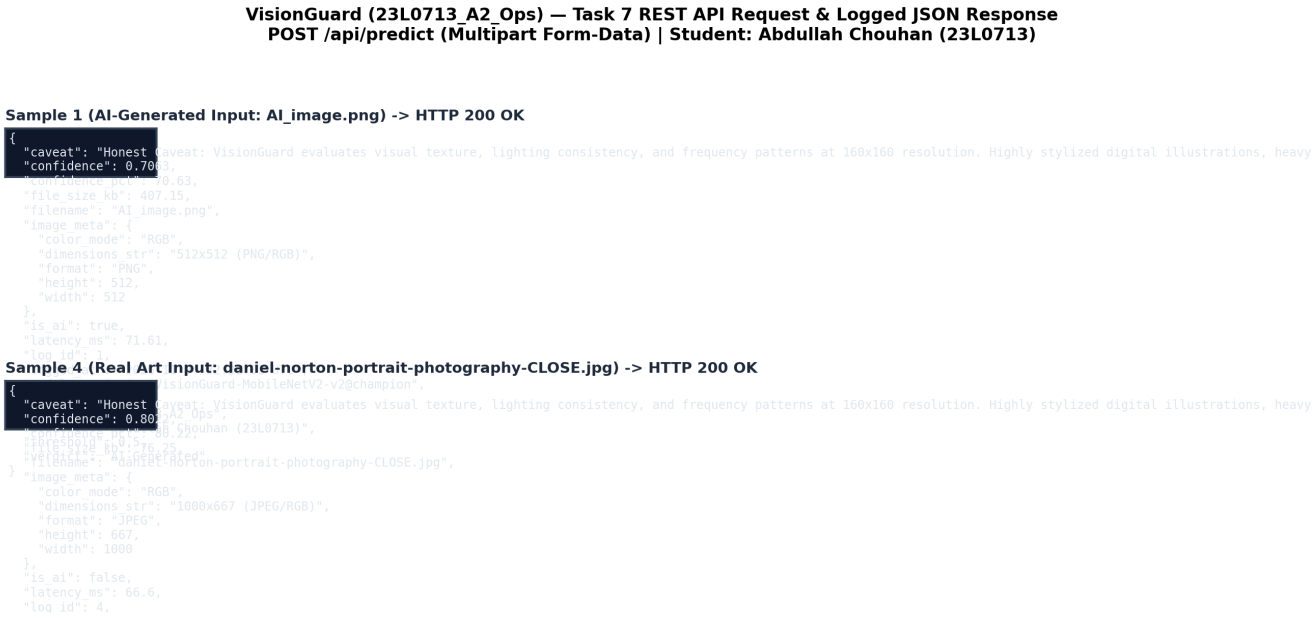

Figure 7.1: REST API POST /api/predict multipart request and logged JSON response payloads.

Task 8.1: Integrating Localhost Flask UI with Cloud API & Database History

How Local Flask Communicates with the Cloud Model API: When USE_CLOUD_API=true and CLOUD_API_URL is set in .env (or selected via the radio toggle on the Flask upload form), the local Flask server (http://localhost:5000) validates the uploaded image in memory and dispatches a multipart POST request with header X-API-Key to {CLOUD_API_URL}/api/predict. It parses the returned JSON verdict, confidence, raw P(AI) score, and model version, records a permanent audit row in prediction_logs with inference_source='cloud_api', and renders both the result page and the /history view.

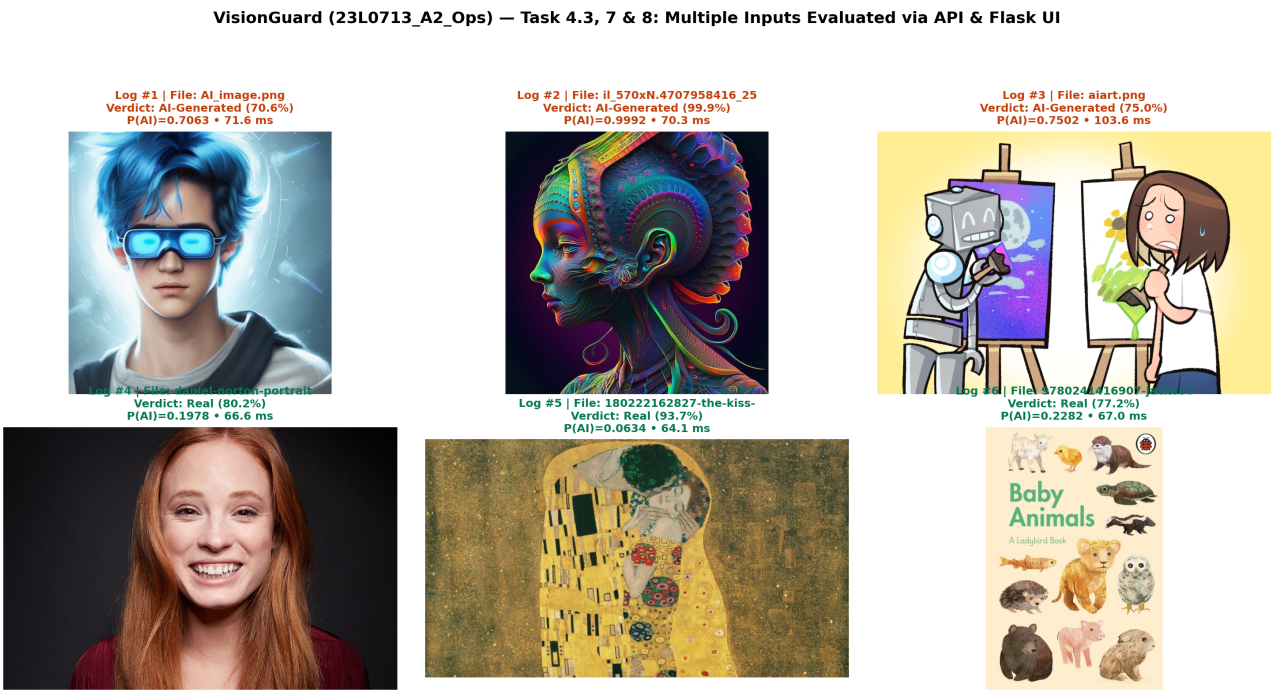

Figure 8.1: Six test inputs (3 AI-Generated, 3 Real) evaluated through the VisionGuard API and logged in the database history trail.